

ZHEZQAZGHAN, KazakhstanWMO: **356710**Lat: **47.800N**Lon: **67.717E**Elev: **346**StdP: **97.24**Time Zone: **6.00 (E06)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.7	-27.0	-33.1	0.2	-29.4	-30.5	0.2	-26.8	12.2	-7.2	10.8	-8.2	2.1	340	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.7	35.9	18.2	34.1	17.6	32.4	17.0	19.3	31.6	18.6	30.7	17.9	29.8	4.7	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.7	11.6	21.9	14.5	10.7	21.7	13.3	9.9	21.9	56.4	31.5	54.1	30.7	51.9	29.8	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.6	10.1	8.7	-32.3	38.9	3.1	1.6	-34.5	40.0	-36.3	41.0	-38.0	41.8	-40.3	43.0		
(5)			-32.7	20.7	3.3	1.3	-35.0	21.6	-36.9	22.4	-38.8	23.1	-41.1	24.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	-13.9	-12.6	-2.6	9.4	16.1	22.0	23.5	22.3	15.3	6.9	-2.5	-10.5		
	DBStd	14.60	6.99	7.02	7.26	5.43	4.30	3.82	3.49	4.07	4.77	5.13	6.09	6.88		
	HDD10.0	2989	742	633	394	74	5	0	0	0	8	123	374	636		
	HDD18.3	4896	1000	866	649	269	96	11	3	13	116	355	624	894		
	CDD10.0	1606	0	0	3	57	193	361	419	380	166	26	1	0		
	CDD18.3	472	0	0	0	2	25	122	164	135	24	1	0	0		
	CDH23.3	6458	0	0	1	39	418	1686	2128	1797	375	14	0	0		
	CDH26.7	2947	0	0	0	4	118	782	1035	870	137	1	0	0		
(14) Wind	WSAvg	3.9	3.6	3.6	4.4	4.4	4.4	4.1	4.1	3.9	3.6	3.6	3.4	3.6		
(15) Precipitation	PrecAvg	190	18	12	17	17	19	15	17	13	11	17	15	24		
	PrecMax	468	46	28	47	59	104	86	46	77	51	44	50	299		
	PrecMin	92	0	1	2	0	0	0	0	0	0	0	2	0		
	PrecStd	80	11	8	10	15	24	18	16	14	14	12	11	51		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.4	2.6	19.0	27.0	32.1	37.0	38.2	37.5	33.9	25.4	13.4	2.9		
		MCWB	0.8	1.3	10.9	14.1	16.0	17.9	19.1	18.8	16.7	12.1	9.8	1.1		
	2%	DB	0.2	0.7	13.7	24.1	29.7	34.7	35.9	35.2	30.3	21.9	10.2	1.0		
		MCWB	-0.3	-0.2	8.3	12.6	15.5	17.7	18.3	17.9	15.1	10.6	6.7	0.0		
	5%	DB	-1.6	-0.6	10.0	21.7	27.7	33.0	34.0	33.5	27.6	19.0	7.5	-0.1		
		MCWB	-2.2	-1.4	6.2	11.3	14.7	17.2	17.7	17.4	14.0	9.5	4.8	-0.9		
	10%	DB	-3.5	-2.1	6.6	19.2	25.6	31.1	32.0	31.7	24.9	16.1	5.2	-1.5		
		MCWB	-4.1	-2.9	4.1	10.2	13.8	16.7	17.2	16.9	13.1	8.5	3.2	-2.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.0	1.5	11.2	15.2	17.8	20.3	20.2	20.3	17.3	13.1	10.9	1.5		
		MCDB	1.3	2.4	18.3	24.3	27.1	31.4	32.5	34.4	32.0	22.4	13.0	2.0		
	2%	WB	-0.3	0.0	8.7	13.6	16.5	19.0	19.4	18.9	15.7	11.5	7.3	0.1		
		MCDB	0.2	0.6	13.0	22.5	26.7	30.8	31.4	32.4	28.4	19.8	9.4	0.7		
	5%	WB	-2.1	-1.3	6.4	12.2	15.4	18.2	18.7	18.0	14.5	10.2	5.0	-1.0		
		MCDB	-1.6	-0.6	9.7	20.0	25.5	29.9	30.5	31.5	25.9	17.6	7.0	-0.2		
	10%	WB	-4.1	-2.9	4.1	10.9	14.5	17.4	18.0	17.2	13.5	8.8	3.3	-2.3		
		MCDB	-3.6	-2.2	6.7	18.0	24.1	28.7	29.5	29.9	24.0	15.4	5.2	-1.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.6	9.3	12.8	14.1	14.1	13.7	14.2	14.6	12.6	8.7	8.4		
		MCDBR	5.9	7.3	13.2	16.3	16.9	16.6	16.2	17.0	17.9	17.3	11.4	7.2		
	5% WB	MCWBR	5.8	7.0	8.3	7.6	7.0	6.0	5.7	6.4	7.8	9.3	8.5	6.5		
		MCWBR	5.9	7.3	12.7	14.8	15.1	14.5	14.4	15.4	16.4	15.6	10.0	6.6		
(40) Clear-Sky Solar Irradiance	taub		0.306	0.342	0.394	0.424	0.434	0.426	0.427	0.397	0.355	0.351	0.337	0.313		
	taud		2.079	1.987	2.046	2.078	2.108	2.199	2.218	2.283	2.353	2.275	2.201	2.128		
	Ebn at Noon		730	783	806	823	831	840	832	843	849	779	695	670		
	Edh at Noon		91	126	141	150	152	140	135	121	102	94	82	78		
(44) All-Sky Solar Radiation	RadAvg		1.55	2.68	3.86	5.22	6.62	7.01	6.68	6.08	4.72	2.79	1.45	1.11		
	RadStd		0.26	0.28	0.38	0.48	0.44	0.41	0.33	0.35	0.18	0.25	0.14	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page